PRISMA

Table S3: PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes, “This systematic review evaluates how…”
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes, Landsat + lentic waters + water-quality parameters + main exclusions
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes, WOS + Scopus + **June 10, 2026**
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Not Applicable
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes, Classification + qualitative synthesis
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes, 229 studies, 1983–2026
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes, Evolution MSS/TM → OLI/OLI-2 + Open Data Policy + ML
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes, identify the principal technical bottlenecks limiting operational implementation
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes, temporal backbone + framework + roadmap
OTHER			
Funding	11	Specify the primary source of funding for the review.	No
Registration	12	Provide the register name and registration number.	No

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TITLE			
Title	1	Identify the report as a systematic review.	Yes
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract (PRISMA 2020 for Abstracts Checklist)
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Introduction
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Introduction
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	**Methods – Literature search and study selection; Table 1; Fig. 1.** Inclusion and exclusion criteria and the criteria used to group studies for synthesis are described in this section and summarized in Table 1 and the PRISMA flow diagram.
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	**Methods – Literature search and study selection.** Web of Science (WOS) and Scopus were searched up to June 10, 2026.
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	**Methods – Literature search and study selection.** Search terms (“inland water quality remote sensing”, and subsequent filtering by “Landsat”), databases searched, and search date are reported.
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	**Methods – Literature search and study selection; Fig. 1.** The screening and full-text eligibility procedures, together with predefined reasons for exclusion at each stage, are described and quantified in the PRISMA flow diagram.
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	**Methods – Data collection and extraction process; Supplementary Material 1.** Data were manually extracted by one reviewer using a predefined extraction framework and subsequently reviewed by the co-authors. Discrepancies were resolved by consensus. Study investigators were not contacted to obtain or confirm missing information, and no automated tools were used for screening or data extraction.
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	**Methods – Data extraction and classification; Supplementary Material 1.** Water-quality parameters, methodological approaches, model-performance metrics, and other variables extracted from the included studies are

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			specified.
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	**Methods – Data extraction and classification; Supplementary Material 1.** Additional variables included geographic location, study period, water-body type, Landsat sensor, spectral information, methodology, and application.
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	**Methods – Literature search and study selection.** A formal risk-of-bias assessment was not performed because the review synthesized methodological and technological studies rather than clinical or intervention-based evidence. Study quality was considered during the screening process by evaluating methodological relevance, completeness of reporting, and consistency with the review objectives.
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	**Methods – Data synthesis.** Not applicable. No quantitative effect measures were calculated because this review presents a qualitative synthesis of technological and methodological developments.
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	**Methods – Data extraction, classification and synthesis; Table 1.** The studies were classified according to: sensor, parameter, methodology, time period, theme as shown in the results
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	**Methods – Data extraction and classification; Supplementary Material 1.** Terminology for water-quality parameters, sensors, methodologies, and applications was standardized and harmonized prior to synthesis. No imputation of missing experimental data was performed.
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	**Methods – Data synthesis; Results; Figs. 4-8; Supplementary Material 1.** Extracted data were tabulated and visually summarized using temporal analyses and graphical comparisons among water-quality parameters, waterbody type, methodological approaches, applications, and geographic regions.
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Studies were synthesized through qualitative thematic analysis according to technological evolution, methodological approaches, monitored water-quality parameters, sensor generations, and future research directions.
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup	Not applicable.

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		analysis, meta-regression).	The variability between studies was interpreted qualitatively considering sensors, regions, parameters, and methodologies
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not applicable.
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not applicable. No formal assessment performed.
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not applicable.
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	See section 2, paragraph 2 and figure 1. PRISMA flow diagram
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	**Methods – Literature search and study selection; Table S2.** Studies that appeared potentially eligible but were excluded following full-text assessment are identified, together with the specific reason for exclusion.
Study characteristics	17	Cite each included study and present its characteristics.	Supplementary Material 1.
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	A formal study-level risk-of-bias assessment was not conducted because the included literature mainly comprised methodological and remote-sensing application studies rather than intervention or effect-estimation studies, for which standardized risk-of-bias tools have been developed. Instead, study quality and relevance were addressed through predefined eligibility criteria and systematic screening.
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Not applicable in its conventional quantitative sense. Individual studies were characterized through standardized extraction of methodological, technological and application-related variables rather than through common effect estimates.
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	For each thematic synthesis, the characteristics of the contributing studies were summarized according to sensor generation, target water-quality variable, retrieval approach, study context and temporal evolution. See 3 to 8 sections
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Not applicable. No meta-analysis or quantitative statistical synthesis was performed.

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	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Heterogeneity was explored qualitatively by comparing differences across Landsat sensors, geographical regions, water-quality variables, optical conditions and retrieval methodologies. See 3 to 8 sections.
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not applicable. No statistical sensitivity analyses were conducted because the review did not involve meta-analysis or pooled effect estimates.
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	No formal assessment of publication or reporting bias was performed because the review did not synthesize comparable quantitative effect estimates.
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not applicable. The certainty of evidence was not assessed using frameworks such as GRADE because the review focuses on technological and methodological evolution rather than intervention effects or clinical outcomes.
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	**Sections 4–8.** The findings are interpreted in the context of previous evidence throughout the thematic discussion, particularly regarding the technological evolution of Landsat sensors, methodological developments in water-quality retrieval, and changes in the applications of Landsat-based inland-water monitoring.
	23b	Discuss any limitations of the evidence included in the review.	**Section “Technical Bottlenecks: The Pixel Challenge in Inland Waters”.** Limitations of the available evidence are discussed, including methodological heterogeneity among studies, differences in atmospheric correction, calibration and validation procedures, variability in reported performance metrics, limited comparability among water-quality parameters and algorithms, and geographical and environmental biases in the available literature.
	23c	Discuss any limitations of the review processes used.	**Methods / Limitations of the review**. Limitations of the review process are acknowledged, including restriction to Web of Science and Scopus, English-language publications, Landsat-focused studies, exclusion of rivers and coastal waters, and the absence of a formal quantitative meta-analysis due to methodological heterogeneity.

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	23d	Discuss implications of the results for practice, policy, and future research.	**Section 8. Roadmap: From Earth Observation to Global Freshwater Intelligence**Implications are discussed for operational water-quality monitoring, integration of Landsat with complementary satellite missions, methodological standardization, development of transferable retrieval models, policy-oriented monitoring, and future research.
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	The review was not prospectively registered.
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	No review protocol was registered for this systematic review.
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not applicable. No formal review protocol was prepared or registered; therefore, there were no amendments to a protocol or registration to report.
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	This research received no external funding.
Competing interests	26	Declare any competing interests of review authors.	The authors declare no competing interests.
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	The complete database of the 229 studies included in this review, together with the standardized data extraction table containing all variables used in the qualitative synthesis, is available as Supplementary Material.

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